

Research Project - Intermediate Report

District-Level Dengue Outbreak Early Warning System for Sri Lanka

Group Number: Group 10

Group Members:

Name	Student ID
Kavindu Amalka	SE/2021/008
Hiruni Imasha	SE/2021/053
Sandaru Mihiran	SE/2021/010
Sachini Weerakkody	SE/2021/018

Date: July 7, 2026

Contents

1	Introduction & Project Overview	2
1.1	Research Objectives	2
2	Comprehensive Literature Review	3
3	Experimental Setup & Research Design	9
3.1	Detailed Methodology Description	9
3.2	Tools, Technologies, and Frameworks Selected	11
3.3	Research Framework	12
3.4	Validation Approach	14
4	Data Collection Progress	16
4.1	Data Sources Identification and Access Portals	16
4.1.1	Epidemiological Data Source: Ministry of Health Epidemiology Unit	16
4.1.2	Meteorological Data Source: NASA POWER API Engine	16
4.2	Data Collection and Engineering Methods Implemented	16
4.2.1	Manual Epidemiological Data Extraction Framework	16
4.2.2	Meteorological Feature Ingestion and API Query Configuration .	17
4.2.3	Spatiotemporal Data Fusion & Longitudinal Panel Dataset Alignment	17
4.3	Sample Dataset Architecture and Documentation	17
4.3.1	Spatial Granularity and Longitudinal Framework	17
4.3.2	Schema Definition and Data Dictionary	17
4.3.3	Missing Value Resolution via Advanced Imputation	17
4.4	Data Acquisition Challenges and Risk Mitigations	18
4.4.1	Institutional Barriers & Privacy Restrictions in Telecommunication Mobility Datasets	19
4.4.2	Financial Procurement Constraints for National Meteorological Data	19
4.4.3	Layout Irregularities and Structural Inconsistencies in Weekly PDF Reports	20
5	Preliminary Results & Discussion	21
5.1	Exploratory Data Analysis and Climate–Dengue Visualizations	21
5.2	District-Specific Temporal Lag Discovery	22
5.3	Preliminary Baseline Model Performance Metrics	24
5.4	Discussion of Early Epidemiological Insights	24

1 Introduction & Project Overview

Dengue fever is a critical, hyperendemic public health crisis in Sri Lanka. Its transmission is closely linked to seasonal monsoonal climate shifts, yet current national management strategies remain reactive, relying on retrospective hospital records rather than anticipation. Consequently, vector control measures are often deployed after outbreaks peak, causing sub-optimal medical resource allocation. To transition from crisis management to proactive strategic prevention, this research introduces an early warning system covering all 26 Regional Director of Health Services (RDHS)/District areas across Sri Lanka. By leveraging advanced statistical, machine learning, and deep learning architectures, the system integrates historical epidemiological trends with micro-climatic indicators to deliver a 4-week-ahead outbreak forecasting index. This early warning buffer empowers Public Health Inspectors (PHIs) to precision-target vector populations and mobilize emergency medical resources before clinical surges occur.

1.1 Research Objectives

- **Data Fusion:** Integrate historical weekly dengue cases extracted from the Epidemiology Unit’s Weekly Epidemiological Reports with synchronized meteorological data (Precipitation, Temperature, Humidity) from the NASA POWER API into a unified 26 RDHS/District areas longitudinal panel dataset.
- **Comparative Modeling Framework:** Implement and evaluate parallel forecasting pipelines utilizing SARIMAX, LightGBM, and LSTM architectures to benchmark statistical, machine learning, and deep learning approaches under a uniform predictive environment.
- **Dual-Stream Prediction Output:** Configure the modeling architectures to deliver dual forecasting streams, generating both continuous numbers for 4-week-ahead case counts (Regression) and localized outbreak risk alerts (Classification).
- **Regional Adaptation & Lag Analysis:** Isolate RDHS/District-specific climate-to-outbreak time lags to capture the unique epidemiological dynamics and seasonal monsoon variations between Sri Lanka’s Wet, Dry, and Intermediate zones.
- **National Scale Generalization:** Expand predictive modeling across all 26 administrative health regions (RDHS)/District simultaneously, overcoming the geographical and data-availability limitations of prior localized studies.
- **Decision-Support Interface:** Build an interactive React.js web dashboard that translates complex modeling forecasts into intuitive geospatial risk visualizations and alert levels for health authorities.

2 Comprehensive Literature Review

Dengue fever remains one of the most persistent vector-borne public-health challenges in tropical and subtropical countries. In Sri Lanka, dengue has become an endemic and periodically epidemic disease, with repeated outbreaks placing pressure on hospitals, public-health officers, vector-control teams, and regional health authorities. The need for early warning is therefore not only a technical forecasting problem but also a public-health decision-support requirement. Existing studies have shown that dengue incidence is influenced by historical case trends, precipitation, temperature, humidity, monsoon seasonality, urbanization, population density, district-level geography, and surveillance practices. However, the challenge is that these factors do not behave uniformly across all locations. Dengue risk in Colombo or Gampaha cannot be assumed to follow the same pattern as Jaffna, Batticaloa, Ratnapura, Nuwara Eliya, or Kalmunai. This makes district/RDHS-level forecasting particularly important for Sri Lanka.

Previous Sri Lankan studies provide strong evidence that dengue is both temporally seasonal and spatially uneven. Fernando et al. (2024) reviewed the epidemiological burden of dengue in Sri Lanka and showed that the disease has remained a major national health concern over multiple years, with high-burden years such as 2017 and 2019. Their review is important because it establishes dengue as a recurring national problem rather than an isolated outbreak event. However, burden-focused studies mainly describe the scale of the disease; they do not by themselves provide a predictive system that can support early intervention. Therefore, while such studies justify the need for dengue early warning, they also reveal the need to move beyond retrospective reporting toward forward-looking prediction.

Spatial studies further show that dengue distribution in Sri Lanka is not uniform. Edirisinghe (2017a) used GIS-based spatial analysis in Matara and demonstrated that dengue risk is concentrated in specific urban and environmental settings rather than being evenly distributed across an area. Similarly, Sun et al. (2017) analysed dengue and climate characteristics across all 25 Sri Lankan districts and identified important hotspot and cold-spot patterns. Their study found that Colombo, Gampaha, and Jaffna were important high-risk areas during the study period and argued that applying the same dengue-control strategy across all districts may not be effective. This finding is directly relevant to the present research because it supports district-level risk modelling and dashboard-based hotspot visualization. However, Sun et al. (2017) used monthly dengue data, which limits the ability to provide short-term weekly early warnings. For practical public-health action, weekly predictions are more useful because field interventions, vector-control activities, and resource allocation are usually planned on shorter operational cycles.

Prabodanie et al. (2020) provide one of the strongest foundations for the district-level

and climate-zone-aware direction of this study. They analysed weekly dengue incidence data for all 25 districts of Sri Lanka from January 2007 to July 2019 and showed that Sri Lanka does not follow a single national dengue pattern. Their findings indicate that southwest wet-zone districts often experience two dengue epidemic waves per year, while northern and eastern dry-zone coastal districts often show one major epidemic wave. They also found stronger synchrony among wet-zone districts, meaning that many districts in the southwest region tend to rise and fall together. This is highly important for the present research because it shows that dengue forecasting should not treat Sri Lanka as one homogeneous epidemiological unit. Instead, prediction models should consider district/RDHS area, wet-zone/dry-zone/intermediate-zone differences, and monsoon-related seasonal patterns. Nevertheless, Prabodanie et al. (2020) focused mainly on spatiotemporal pattern analysis rather than building a predictive early warning model or dashboard. Therefore, their work strongly supports the design of the current research but leaves room for predictive modelling and decision-support implementation.

A more recent systematic review by Wickramasinghe et al. (2026) also confirms the geographical limitations of many Sri Lankan dengue studies. Their review found that many spatial dengue studies are concentrated on one or a few districts, with Colombo and Gampaha receiving the most attention, while many other districts remain underrepresented. The review also reported that only a limited number of studies considered the whole country and that many studies used relatively short time periods. Importantly, the review highlighted that the same risk factor can behave differently across districts. This supports the concept of spatial non-stationarity, where precipitation, humidity, urban land use, or population density may not have the same effect everywhere. For the current research, this justifies the use of a district/RDHS-week panel dataset rather than a single national time series. It also supports the inclusion of district-level features, climate-zone indicators, and monsoon-period variables. However, Wickramasinghe et al. (2026) focus on spatial distribution studies rather than developing a practical forecasting framework. Therefore, their review identifies a gap that the present research directly addresses: the need for district-level predictive early warning tools that combine spatial, temporal, and climatic information.

Climate is one of the most frequently studied drivers of dengue transmission. Precipitation can create mosquito breeding sites, humidity can support mosquito survival, and temperature can influence mosquito development and viral transmission. However, the relationship between weather and dengue is rarely immediate. There is usually a delay between precipitation or humidity changes and reported dengue cases because water accumulation, mosquito breeding, vector development, human infection, incubation, healthcare seeking, and reporting all take time. Edirisinghe (2017b) examined precipitation patterns and dengue epidemics in Sri Lanka and showed that dengue incidence may increase several weeks after precipitation begins. This supports the idea that precipitation

should not only be used as a current-week variable but also as a lagged predictor. Withanage et al. (2018) strengthened this argument through a Gampaha-based forecasting model that used lagged precipitation, rainy days, minimum temperature, minimum relative humidity, and previous dengue incidence. Their model demonstrated the practical value of weather-lag variables for early warning. However, the study was district-specific and used monthly-level modelling, so its findings cannot be directly generalized across all Sri Lankan districts or RDHS areas.

Uduwanage et al. (2025) made an important contribution by comparing CNN, XGBoost, and SARIMAX for dengue outbreak prediction in selected Sri Lankan districts using weather variables. Their study found that same-month weather correlations were weak but improved after applying time-shifted weather variables, supporting the biological logic of delayed mosquito-life-cycle effects. Their model comparison reported SARIMAX as the best-performing model among the tested approaches, with a lower RMSE than XGBoost and CNN. This is one of the main reasons SARIMAX is selected as the statistical baseline in the present study. However, Uduwanage et al. (2025) also had limitations that are directly relevant to the present research gap. Their weather data came from the Sri Lanka Department of Meteorology and was only available for selected districts, causing some areas to be excluded. Their weekly dengue records were also converted into monthly summaries. Therefore, although their work strongly supports SARIMAX and lagged weather effects, it also leaves an opportunity for a broader district/RDHS-level weekly framework using a more consistently available weather source such as NASA POWER.

Traditional statistical models such as ARIMA, ARIMAX, and SARIMAX are widely used in dengue forecasting because dengue case data is time-dependent. Karasinghe et al. (2024) developed a modified ARIMA model for weekly dengue incidence forecasting in Colombo District. Their final ARIMA(2,1,0) + AR(16) model showed strong training performance, but validation performance weakened during the COVID-19 affected period. This study is useful because it proves the value of weekly dengue forecasting in Sri Lanka. At the same time, it shows the limitation of relying only on historical case values. ARIMA does not directly include precipitation, temperature, humidity, monsoon period, or district-level climate-zone differences. ARIMAX improves this by including external variables, as seen in Sun et al. (2017), but SARIMAX is more suitable for the current study because it can represent both seasonality and external weather variables. Therefore, SARIMAX is not selected simply because it was used before; it is selected because it is the most appropriate Sri Lankan literature-supported baseline for a dengue forecasting problem involving time series, seasonality, and weather variables.

International dengue forecasting studies show that machine learning and deep learning can add value, but they must be locally validated before being applied to Sri Lanka. Rahman et al. (2025) developed an interpretable tree-based dengue early warning model

in Bangladesh using Random Forest, XGBoost, and LightGBM with dengue surveillance, climate, sociodemographic, and landscape variables. LightGBM achieved the best overall performance among the tested models, and SHAP analysis was used to explain important predictors. This is relevant because the present research also uses a structured tabular dataset containing dengue lags, precipitation lags, humidity lags, temperature lags, epidemiological week, climate zone, and monsoon-period features. LightGBM is suitable for such data because it can capture nonlinear relationships between these variables. However, Rahman et al. (2025) was conducted in Bangladesh, and its exact climate thresholds, feature effects, and performance values cannot be directly transferred to Sri Lanka. Thus, the present research uses Rahman et al. (2025) as methodological support for selecting LightGBM and explainability methods, but the model must be trained and evaluated using Sri Lankan district/RDHS-level data.

Deep learning studies provide another useful direction, especially for sequence forecasting. Chen and Moraga (2026) used LSTM models for dengue forecasting and outbreak detection in Brazil and showed that adding climate and mobility variables improved performance compared with case-only models. Their work is relevant because the present research also uses a four-week-ahead forecasting horizon and treats dengue prediction as a temporal sequence problem. However, Chen and Moraga (2026) included mobility data, while the current study focuses only on weekly dengue cases and NASA POWER weather variables due to data availability and feasibility. This difference is important because the present research should not overclaim mobility-aware forecasting. Instead, LSTM is included as a deep learning comparison model to test whether sequence learning from dengue and weather histories improves four-week-ahead prediction under Sri Lankan conditions. The inclusion of LSTM is therefore exploratory and comparative, not based on the assumption that deep learning will automatically outperform SARIMAX or LightGBM.

Other international studies also show the value of climate-aware and interpretable dengue forecasting. Zhang et al. (2026) studied dengue and climate relationships in Singapore and showed that climate effects can appear through different temporal frequencies and lag structures. Al Bannoud et al. (2026) examined machine learning robustness under extreme dengue outbreak conditions in São Paulo and showed that boosted tree models and explainability methods can support outbreak forecasting. Del-Aguila-Mejía et al. (2024) used system dynamics and risk mapping in Spain, demonstrating the value of climate- and vector-aware risk assessment even in a non-endemic context. Feng et al. (2024) reviewed One Health dengue control in China and emphasized that effective dengue prevention requires early warning, surveillance, multisectoral coordination, and rapid response. These studies support the broader direction of the present research: dengue early warning should not stop at numerical prediction but should connect data, modelling, interpretation, and public-health decision-making.

Operational decision support remains a major limitation in dengue forecasting literature. Many studies develop models, report error metrics, or identify hotspots, but fewer translate results into a usable dashboard or public-health tool. Rahulan et al. (2024), in their case study of dengue control in Gampaha, identified poor use of existing data and weak translation of data into evidence-based action as important operational problems. This finding is crucial because it shows that the gap is not only about prediction accuracy. Even if a model performs well, it will have limited value if public-health officers cannot interpret or use the output. The present research therefore includes a dashboard prototype to display predicted case counts, alert status, dengue trends, weather trends, and, if possible, contributing weather-lag factors. This positions the study as a software engineering and public-health decision-support contribution, not merely a model comparison exercise.

A critical synthesis of the reviewed literature reveals several unresolved gaps. First, Sri Lankan studies have produced valuable findings, but many are limited by geographical scope, often focusing on Colombo, Gampaha, Matara, or selected districts. Second, several studies use monthly data, even though weekly data is more appropriate for short-term outbreak warning. Third, station-based weather data creates availability and representativeness problems; selected weather stations may not represent the full district, and missing weather data can exclude districts from analysis. Fourth, although climate lag is repeatedly identified as important, systematic testing of multiple weather lags across district/RDHS areas remains limited. Fifth, many studies do not compare statistical, machine learning, and deep learning approaches under the same Sri Lankan dataset and prediction horizon. Sixth, prediction outputs are often not integrated into a practical dashboard, limiting their usefulness for field-level decision-making.

The present research is designed to address these gaps in a focused and feasible way. It does not claim to include every possible dengue driver. Variables such as human mobility, PHI larval indices, sanitation, waste management, socioeconomic status, and vector surveillance are important, but they are not consistently available for the current implementation. Therefore, the present study deliberately focuses on the most accessible and consistently obtainable inputs: weekly dengue case data and NASA POWER weather data, specifically precipitation, temperature, and humidity. This scope is narrower than some ideal dengue models, but it is more practical and reproducible for a district/RDHS-level early warning prototype in Sri Lanka.

The novelty of this research is therefore not the invention of a new algorithm, but the integration of existing evidence into a locally relevant early warning framework. The study uses a district/RDHS-level dataset covering 25 districts plus Kalmunai as an additional administrative unit, integrates weekly dengue case data with NASA POWER weather variables, performs one-week to twelve-week climate-lag analysis, incorporates wet-zone/dry-zone/intermediate-zone and monsoon-aware features, compares SARIMAX,

LightGBM, and LSTM under the same four-week-ahead forecasting setup, and produces both regression and classification outputs for dashboard-based decision support. This combination is different from prior Sri Lankan studies that focus on single districts, monthly data, spatial analysis without prediction, or one model family only.

Overall, the literature supports the need for a Sri Lanka-specific, climate-aware, district/RDHS-level dengue early warning framework. Sri Lankan studies justify the importance of dengue burden, spatial variation, precipitation effects, monsoon timing, and SARIMAX-based forecasting. International studies justify the inclusion of LightGBM, LSTM, SHAP-style interpretability, and dashboard-oriented early warning ideas. However, the reviewed evidence also shows that these components have not yet been sufficiently integrated into one practical Sri Lankan framework using weekly dengue data and consistent NASA POWER weather inputs. The current research is positioned to fill this gap by developing a four-week-ahead dengue early warning system that connects data preparation, climate-lag analysis, feature engineering, comparative modelling, validation, and dashboard-based public-health interpretation.

3 Experimental Setup & Research Design

3.1 Detailed Methodology Description

This research follows a quantitative experimental research design combined with a software prototype development approach. The main objective is to develop a district/RDHS-level dengue early warning framework for Sri Lanka using weekly dengue case data and NASA POWER weather data. The system is designed to predict dengue risk four weeks ahead, so that public-health officers can identify possible outbreak situations before they become severe.

The first phase of the methodology focuses on data collection from two primary sources. Weekly district/RDHS-level dengue case data is obtained from the Weekly Epidemiological Reports published by the Epidemiology Unit, Ministry of Health, Sri Lanka. This dataset provides historical dengue incidence counts, which are used as the target variable in the study. The second source is meteorological data retrieved from the NASA POWER API using representative geographic coordinates for all 25 districts of Sri Lanka, along with Kalmunai as an additional administrative unit. The key weather variables considered include precipitation, temperature, and relative humidity. To automate data retrieval, a Python-based web application was developed using the Streamlit framework, enabling users to select a custom date range and extract weather data for all districts simultaneously. Since NASA POWER provides daily-resolution data, the system aggregates these values into weekly summaries using the Pandas library, ensuring compatibility with epidemiological reporting. The resulting weather data is then aligned with dengue case records based on district/RDHS area and epidemiological week.

The second phase involves data preprocessing and integration to construct a unified analytical dataset. Weather data is standardized by formatting district names, calendar years, dates, and epidemiological week identifiers to ensure consistency across all records. Daily meteorological observations are converted into weekly aggregates, where precipitation is represented as total weekly precipitation, while temperature and humidity are computed as weekly average values. The dengue and weather datasets are then merged using common fields such as district/RDHS area, year, and epidemiological week. The final output is a structured district/RDHS-week panel dataset, where each record represents a specific geographic area in a given epidemiological week, containing both dengue incidence and corresponding weather conditions.

The third phase of the study focuses on exploratory data analysis (EDA) and climate-lag analysis. Climate-lag analysis is used to identify the delayed effect of weather conditions on dengue transmission, since changes in precipitation, temperature, and humidity do not result in immediate increases in dengue cases. Instead, there is a biological and epidemiological delay due to mosquito breeding cycles, vector development, and reporting

time. To capture this relationship, different lag periods ranging from 1-week to 12-weeks are tested to determine which weather conditions show the strongest delayed association with dengue incidence across districts and, where applicable, across climate zones such as wet, dry, and intermediate zones. In addition, exploratory analysis is performed using Power BI to visualize temporal and spatial patterns in the dataset. Time-series visualizations, such as the weekly dengue trend chart, are generated to observe outbreak peaks, seasonal fluctuations, and long-term variations in dengue incidence, supporting the identification of epidemic periods and potential weather–disease relationships.

The fourth stage is feature engineering. Feature engineering means creating additional model input variables from the original dengue and weather data. These features help the models learn delayed and seasonal dengue patterns. Dengue history features may include previous week cases, two-week lag cases, four-week lag cases, and rolling average case counts. Weather features may include precipitation lag, temperature lag, humidity lag, and rolling weather averages. A rolling average means the average value over recent weeks, such as the average dengue cases during the previous four weeks. Seasonal features such as epidemiological week, month, Southwest monsoon, Northeast monsoon, and inter-monsoon periods will also be added. These features help the models learn both short-term and seasonal dengue behaviour.

The fifth stage is comparative modelling. This research compares three representative modelling approaches: SARIMAX, LightGBM, and LSTM. SARIMAX is selected as the main statistical baseline because it can model time-series seasonality and external weather variables, and it has shown strong performance in Sri Lankan dengue forecasting literature. LightGBM is selected as the machine learning model because it can handle nonlinear relationships in tabular data, such as dengue lags, weather lags, monsoon indicators, and district-level features. LSTM is selected as the deep learning model because it can learn sequential patterns from previous weeks of dengue and weather data.

The modelling output follows a dual-stream prediction design. The first stream is a regression stream, where the model predicts the expected number of dengue cases four weeks ahead. The second stream is a classification stream, where the system predicts whether a district/RDHS area is likely to enter an outbreak or high-risk condition four weeks ahead. For classification, a district-specific alert threshold can be created using historical dengue case patterns, such as historical mean plus two standard deviations. This allows the system to provide both a numerical forecast and a practical warning decision.

After model training, the models will be evaluated using regression and classification metrics to identify the most suitable model for dashboard integration. This phase focuses on comparing different models based on their predictive performance and selecting the best-performing model for deployment.

The final stage is dashboard prototype development. The dashboard will mainly

display predicted dengue case counts, alert status, dengue trends, and weather trends. If implementation time allows, district risk maps and main contributing weather-lag factors will also be included. This makes the research useful not only as a prediction experiment but also as a public-health decision-support system.

3.2 Tools, Technologies, and Frameworks Selected

The tools and technologies selected for this research are chosen based on data processing, modelling, evaluation, explainability, and dashboard development requirements.

Table 1: Selected Tools, Technologies, and Frameworks

Category	Selected Tools / Technologies	Purpose
Programming language	Python	Main language for data processing, modelling, and evaluation.
Data handling	Pandas, NumPy	Preparing the weather dataset, integrating dengue case values, generating features, and analysing dengue-weather patterns.
Weather data source	NASA POWER API	Collecting precipitation, temperature, and humidity values using representative district/RDHS coordinates.
Statistical modelling	statsmodels / pmdarima	Developing and testing SARIMAX time-series models.
Machine learning	LightGBM, scikit-learn	Training LightGBM models and supporting preprocessing, validation, and metric calculation.
Deep learning	TensorFlow/Keras or PyTorch	Developing LSTM-based time-series models to learn sequential dengue-weather patterns.
Visualization	Power BI, Matplotlib, Plotly	Creating dengue trend graphs, weather graphs, lag-analysis plots, model result visualizations, and interactive dashboard charts.
Explainability	SHAP / Feature importance	Explaining which features contribute most to dengue predictions, especially for the LightGBM model.
Model evaluation	scikit-learn metrics, NumPy, Pandas	Calculating MAE, RMSE, MAPE, Precision, Recall, F1-score, and ROC-AUC.
Development	Streamlit, React	Building the early warning dashboard prototype. Streamlit is suitable for fast Python-based dashboard development, while React may be considered later for a more customized frontend.
Version control	GitHub	Managing source code, documentation, experiment versions, and collaboration.

The study uses Python as the primary programming language due to its strong support for data processing, time-series modelling, machine learning, and deep learning. Data handling is performed using Pandas and NumPy to prepare the weather dataset, integrate dengue case data, generate features, and analyse dengue–weather relationships. NASA POWER API is used as the weather data source to collect precipitation, temperature, and humidity values based on district/RDHS coordinates.

Statistical modelling is carried out using statsmodels or pmdarima for SARIMAX-based time-series forecasting, while LightGBM and scikit-learn are used for machine learning model development, preprocessing, validation, and performance evaluation. Deep learning models are implemented using TensorFlow/Keras or PyTorch to capture sequential dengue–weather patterns.

For visualization, Power BI, Matplotlib and Plotly are used to generate dengue trends, weather patterns, lag-analysis results, and model outputs. Model interpretability is achieved using SHAP and feature importance techniques, particularly for the LightGBM model, to identify key influencing variables.

Model evaluation is performed using scikit-learn metrics along with NumPy and Pandas to compute performance measures such as MAE, RMSE, MAPE, Precision, Recall, F1-score, and ROC-AUC.

For system development, Streamlit and React are used to build the early warning dashboard prototype, where Streamlit has already been implemented for weather data handling and will be extended for dashboard development. GitHub is used for version control, supporting code management, documentation, and collaboration throughout the project.

3.3 Research Framework

The proposed research framework contains seven main stages: data acquisition, data preprocessing and integration, exploratory data analysis and climate-lag analysis, feature engineering, comparative modelling, validation and evaluation, and dashboard prototype development.

The framework is designed to support four-week-ahead dengue early warning. The input layer contains weekly dengue case data and NASA POWER weather data. The preprocessing layer converts all data into a common district/RDHS-week format. The analysis layer studies dengue trends, weather patterns, wet-zone and dry-zone differences, and delayed climate effects. The feature engineering layer creates model-ready variables such as dengue lags, weather lags, rolling averages, monsoon indicators, and climate-zone indicators.

The modelling layer compares SARIMAX, LightGBM, and LSTM under the same prediction setup. SARIMAX acts as the statistical baseline model. LightGBM acts as

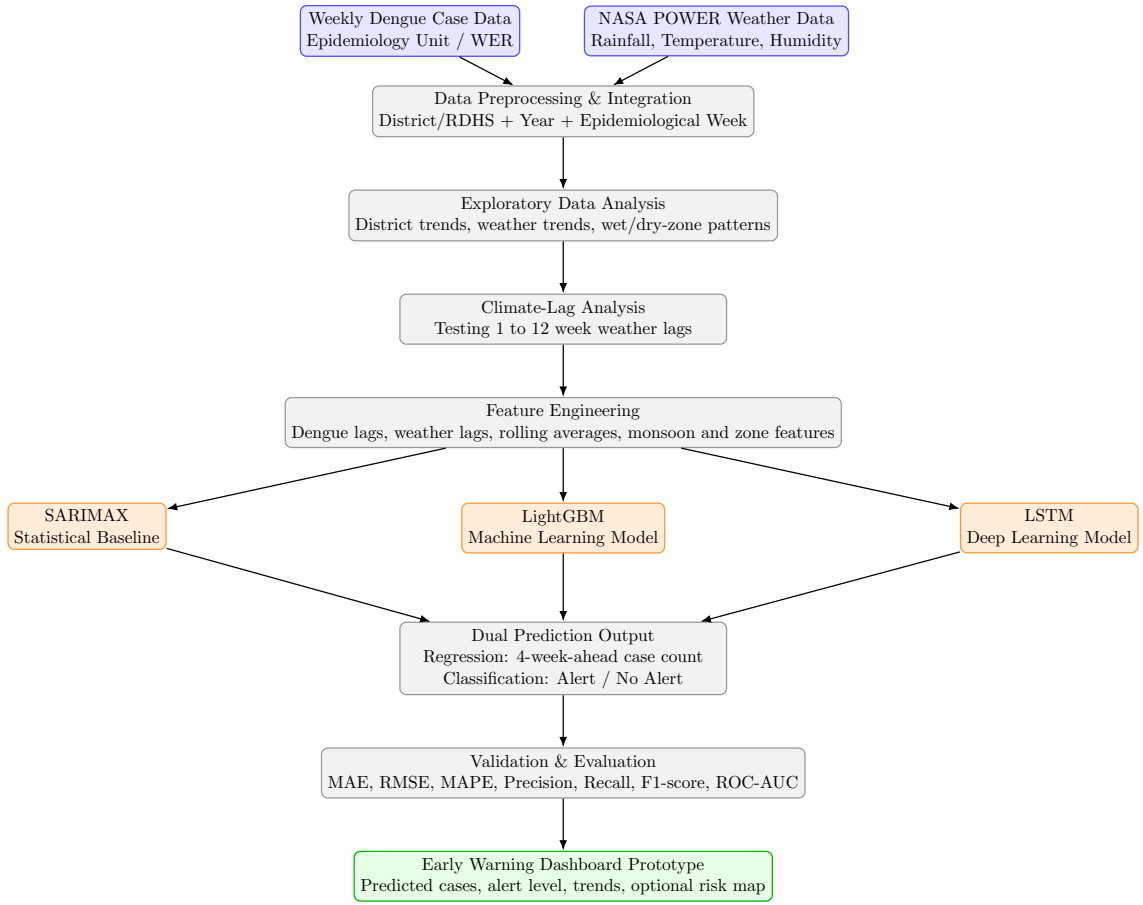


Figure 1: Proposed dengue early warning research framework

the main machine learning model for structured lag-based features. LSTM acts as the deep learning sequence model. This comparison allows the research to evaluate whether statistical, machine learning, or deep learning approaches are more suitable for Sri Lankan district/RDHS-level dengue early warning using NASA POWER weather data.

The output layer produces both predicted case counts and outbreak alerts. The validation and evaluation layer checks whether the models are reliable before integrating the selected model into the dashboard. Finally, the dashboard layer presents the results in a practical format for decision-making.

3.4 Validation Approach

The validation approach is designed to test the models under realistic forecasting conditions. Since dengue forecasting is a time-series problem, random train-test splitting is not suitable. Random splitting can mix past and future data, which may give unrealistic performance. Therefore, this research will use walk-forward validation.

In walk-forward validation, the model is trained using past data and tested on future data. This better represents a real early warning system because the model must always predict future dengue risk using only information available up to the current week.

An example validation setup is shown below:

Training Period	Testing Period
2016–2020	2021
2016–2021	2022
2016–2022	2023
2016–2023	2024

All models will be evaluated using the same dataset, same prediction horizon, and same validation method. This ensures a fair comparison between SARIMAX, LightGBM, and LSTM. Before validation, the dataset will be checked to ensure that dengue cases are correctly aligned with the same district/RDHS area and epidemiological week as the weather variables.

For the regression stream, the models will be evaluated using the metrics shown in Table 2.

For the classification stream, the models will be evaluated using the metrics shown in Table 3.

In outbreak prediction, recall is especially important because missing a real outbreak can be more harmful than issuing a false warning. Therefore, the final model selection will consider not only overall accuracy but also the model’s ability to detect high-risk dengue situations early.

Table 2: Regression Evaluation Metrics

Metric	Purpose
MAE	Measures the average size of prediction errors.
RMSE	Penalizes larger prediction errors more strongly.
MAPE	Shows prediction error as a percentage.
Forecast bias	Shows whether the model usually overpredicts or underpredicts.

Table 3: Classification Evaluation Metrics

Metric	Purpose
Precision	Measures how many predicted alerts were actually correct.
Recall	Measures how many real outbreak situations were detected.
F1-score	Balances precision and recall.
ROC-AUC	Measures the model’s ability to separate alert and non-alert situations.

The final selected model will be chosen based on both prediction performance and practical usefulness. A model with slightly lower numerical accuracy but better outbreak detection and interpretability may be more useful for a public-health early warning system. Therefore, the validation process will consider regression accuracy, classification performance, alert reliability, and interpretability before selecting the final model for dashboard integration.

4 Data Collection Progress

4.1 Data Sources Identification and Access Portals

To engineer a highly resilient and statistically sound predictive framework, it was imperative to construct a multi-dimensional, long-term longitudinal dataset covering a 10-year historical window from January 2016 to May 31, 2026. The data ingestion phase focuses on capturing synchronized epidemiological and micro-climatic feature streams across all 26 Regional Director of Health Services (RDHS)/District areas in Sri Lanka.

4.1.1 Epidemiological Data Source: Ministry of Health Epidemiology Unit

The primary ground-truth source for historical dengue incidence was official national surveillance records compiled by the Ministry of Health, Sri Lanka. Data was systematically accessed and downloaded from the official repository (<https://www.epid.gov.lk/weekly-epidemiological-report/weekly-epidemiological-report>). Retrieved individual Weekly Epidemiological Report (WER) documents in PDF format were used, and for each Epi-Week, data was mapped against the 26 RDHS/District administrative health domains.

4.1.2 Meteorological Data Source: NASA POWER API Engine

Climate metrics were sourced using the NASA POWER (Prediction Of Worldwide Energy Resources) Data Access Viewer API. To match meteorological variables with the 26 RDHS/District health boundaries, the research calculated the unique geographical centroid coordinates (Latitude and Longitude) for each area. These were passed as localized spatial parameters into the API configuration. The engine fetched daily precipitation, relative humidity, and atmospheric temperature, which were subsequently synthesized into uniform weekly climate feature blocks.

4.2 Data Collection and Engineering Methods Implemented

4.2.1 Manual Epidemiological Data Extraction Framework

Due to structural shifts and layout variations in WER PDF documents over the 10-year span, automated parsers introduced high tokenization error rates. To guarantee structural integrity, a rigorous manual extraction methodology was executed. Over 540 weekly PDF documents were individually reviewed, isolated into text-based repositories, and structured using spreadsheet-driven Text-to-Columns boundaries, indexed by Year, Epi-Week, and the 26 RDHS/District administrative domains.

4.2.2 Meteorological Feature Ingestion and API Query Configuration

A custom Python-based web application was architected using the Streamlit framework to automate communication with the NASA POWER API. Utilizing the Python requests library, the application dispatches parallel geospatial queries. Because the NASA POWER infrastructure supplies metrics at a daily resolution, a data engineering pipeline was built within the Pandas library to align the temporal scales. Daily observations were grouped into standardized 7-day epidemiological calendar windows. The aggregation logic computed:

- Weekly mathematical Average (Mean) for Atmospheric Temperature.
- Weekly mathematical Average (Mean) for Relative Humidity.
- A dedicated, cumulative feature column representing the Total Weekly Precipitation (summation of the 7-day precipitation block).

4.2.3 Spatiotemporal Data Fusion & Longitudinal Panel Dataset Alignment

A composite spatiotemporal key was established to join independent epidemiological and meteorological matrices, matching data rows based on a synchronized intersection of RDHS Area Name, Epi-Week Number, and Year. The finalized dataset forms a multi-variable panel configuration containing over a decade of continuous weekly observations.

4.3 Sample Dataset Architecture and Documentation

4.3.1 Spatial Granularity and Longitudinal Framework

The dataset is a balanced longitudinal panel matrix. For every distinct Epi-Week within the 10-year horizon, the database instantiates exactly 26 independent rows, ensuring that the sequential machine learning models (LSTM) capture concurrent, parallel spatial-temporal interactions.

4.3.2 Schema Definition and Data Dictionary

The master Excel database utilizes a standardized tabular schema containing thirteen explicit feature columns. The precise feature vector parameters are detailed in the data dictionary table below:

4.3.3 Missing Value Resolution via Advanced Imputation

During the manual compilation of historical epidemiological data from the Weekly Epidemiological Reports, data fragmentation anomalies were identified due to localized reporting gaps or institutional archiving failures within specific historical years.

Table 4: Data Dictionary for the Master Registry

Column Name	Data Type	Description / Operational Metric
Year	Numerical (Integer)	The calendar year of the recorded observation (2016 – 2026).
Month	Text (String)	The calendar month for the surveillance week.
Week No	Numerical (Integer)	A continuous week counter over the 10-year timeline for LSTM sequence indexing.
Epi Week	Numerical (Integer)	Standardized Epidemiological Week index from the Ministry of Health calendar.
Week Start	Date (YYYY-MM-DD)	The start date of the surveillance week.
Week End	Date (YYYY-MM-DD)	The end date of the surveillance week.
District	Text (String)	Primary district name as text (e.g., Colombo).
Zone	Text (String)	Climatic zone: Wet, Dry, or Intermediate.
Temp C	Numerical (Float)	Weekly mean temperature in °C from NASA POWER.
Humidity Precip mm	Numerical (Float)	Mean daily precipitation over the 7-day window.
Precipitation Total mm	Numerical (Float)	Total weekly precipitation in mm, summed over 7 days.
Dengue Cases	Numerical (Integer)	Total reported weekly dengue cases.

- **Handling Strategy:** Leaving these fields vacant as null strings or setting them to zero would corrupt the sequential longitudinal dependency vectors required by the LSTM architectures. Dropping the rows would similarly compromise the chronological consistency of the time series.
- **Imputation Protocol:** To preserve structural continuity, a Linear Interpolation Framework was adopted using the Python Pandas library. Missing values in the Dengue Cases column were mathematically imputed based on the linear trend line bound between the closest preceding and succeeding valid data boundaries within that specific RDHS zone. This advanced statistical masking ensures an uninterrupted temporal matrix ready for model ingestion.

4.4 Data Acquisition Challenges and Risk Mitigations

Constructing a highly precise, long-term longitudinal framework across heterogeneous data environments presented significant institutional, financial, and structural bottlenecks. This section systematically details the real-world challenges encountered during the data engineering phase and the strategic risk mitigations implemented to ensure project continuity.

4.4.1 Institutional Barriers & Privacy Restrictions in Telecommunication Mobility Datasets

Our initial research strategy aimed to fuse human mobility vectors derived from mobile network telemetry to model inter-district vector propagation.

- **The Challenge:** To execute this, the team initiated formal communication channels and direct physical consultations with major national telecommunication service providers, including Dialog Axiata, Mobitel, and Sri Lanka Telecom (SLT). However, these attempts were ultimately unsuccessful due to rigid corporate data governance frameworks, proprietary security constraints, and strict national data protection laws that prohibit the external release of raw subscriber call detail records (CDRs) or location logs.
- **The Mitigation:** To resolve this roadblock without compromising the analytical depth of the study, a formal scoping realignment was approved by the project supervisor. The human mobility indicator was officially de-scoped from the framework. To offset this reduction in variables, the geographical scope of the machine learning pipeline was scaled up from a localized regional footprint to cover all 26 RDHS/District administrative health zones simultaneously, shifting the predictive focus toward high-resolution spatiotemporal climate indicators.

4.4.2 Financial Procurement Constraints for National Meteorological Data

To develop a predictive framework capable of identifying long-term epidemiological trends, continuous daily and weekly historical climate vectors were required since January 2016.

- **The Challenge:** The team formally explored the acquisition of official meteorological records from the Department of Meteorology, Sri Lanka. However, obtaining continuous, localized panel data for precipitation, temperature, and humidity across all target administrative regions involved highly prohibitive financial procurement costs that vastly exceeded our undergraduate research budgetary limitations.
- **The Mitigation:** To bypass this financial constraint while preserving scientific validity, alternative cloud-based earth observation systems were extensively researched. The team successfully engineered an automated ingestion pipeline utilizing the NASA POWER API infrastructure (funded by the United States Government). Because these datasets are generated through verified global satellite data assimilation models and global grid interpolations, they provided peer-reviewed, high-resolution climatic indicators for our specific 26 RDHS coordinate boundaries completely free of cost.

4.4.3 Layout Irregularities and Structural Inconsistencies in Weekly PDF Reports

The extraction of weekly dengue infection metrics directly from the Weekly Epidemiological Reports (WER) compiled by the Ministry of Health presented severe syntactic layout challenges.

- **The Challenge:** Spanning over a decade, these historical source documents were published in static PDF formats with frequent structural variations. Automated digital parsing tools and text-extraction scripts continuously failed or introduced extreme tokenization errors due to shifting table coordinates, overlapping column boundaries, dual-language string formatting, and inconsistent scanning qualities across different annual archives.
- **The Mitigation:** To enforce absolute data accuracy—which is paramount for training deep learning sequences like LSTMs—the team discarded volatile automated parsing scripts in favor of a rigorous manual pipeline. Every single weekly document was individually processed using a strict Delimiter Copy-Paste Framework. The raw tabular segments were isolated, extracted into plaintext editors to preserve tab-separated boundaries, compiled systematically into spreadsheet environments, and parsed cleanly using native 'Text to Columns' spacing parameters to construct an error-free ground-truth master database.

5 Preliminary Results & Discussion

Following the successful execution of the spatiotemporal data fusion phase, exploratory data analysis (EDA) was conducted using advanced business intelligence frameworks to uncover latent epidemiological signals. This section provides a comprehensive structural analysis of the initial insights and baseline formulations derived from the continuous longitudinal panel dataset (January 2016 – May 31, 2026) spanning the 26 RDHS/District zones.

5.1 Exploratory Data Analysis and Climate–Dengue Visualizations

To evaluate the macro-trends embedded within the structured data core, historical time-series distributions and agro-ecological proportions were mapped directly from the master database.



Figure 2: Sum of Dengue Cases by Week Number

- **Epidemic Macro-Surge Tracking:** As evidenced by the continuous line visualization (“Sum of Dengue Cases by Week_No”), the historical baseline is characterized by intense, highly non-linear epidemic waves rather than stationary fluctuations. A catastrophic hyper-spike is clearly documented between Week_No 70 and 90, where aggregate national weekly cases breached the historical ceiling of 10,000 cases per week. A secondary major outbreak pattern is visible around Week_No 200, sustaining infection counts near 5,000 cases. This structural volatility mathematically justifies the implementation of recurrent deep learning frameworks (LSTM) over traditional autoregressive forecasting configurations.

Sum of Dengue Cases by Zone

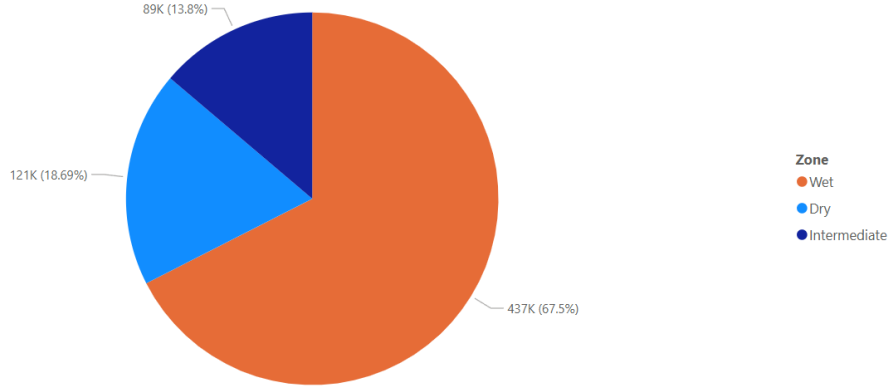


Figure 3: Sum of Dengue Cases by Zone

- Agro-Ecological Spatial Density:** Cross-sectional aggregation mapped via the macro-zonal distribution framework (“Sum of Dengue Cases by Zone”) establishes a stark spatial concentration. The Wet Zone acts as the primary hyper-endemic engine of the island, accounting for an overwhelming 67.5% (437,000 cases) of the cumulative caseload. Conversely, the Dry Zone and Intermediate Zone exhibit lower aggregate footprints, representing 18.69% (121,000 cases) and 13.8% (89,000 cases) respectively. This statistical distribution proves that distinct spatial weight coefficients must be assigned based on environmental zones during model feature engineering.

5.2 District-Specific Temporal Lag Discovery

A vital objective of the exploratory analysis was validating the lag period between micro-climatic triggers and actual recorded hospital admissions across the distinct administrative boundaries.

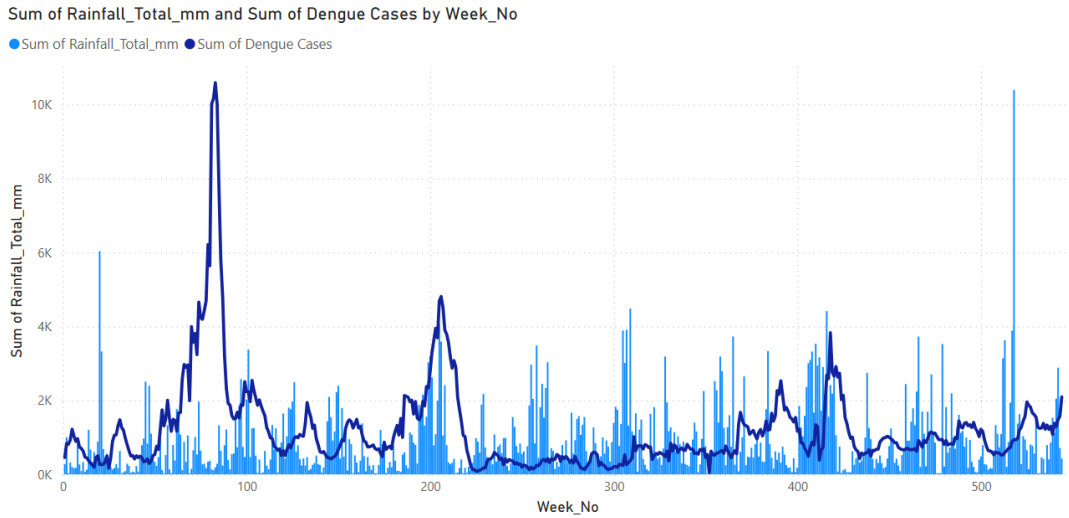


Figure 4: Sum of Precipitation.Total.mm and Sum of Dengue Cases by Week_No

- Precipitation-Driven Larval Ingress (Precipitation Lag):** The dual-axis overlay chart (plotting “Sum of Precipitation_Total_mm” against “Sum of Dengue Cases by Week_No”) visually confirms the biological delay mechanism of the vector. Major monsoonal precipitation events—visible as sharp vertical blue bars—do not result in instantaneous clinical spikes. Instead, a clear, repeatable temporal lag of 3 to 5 weeks is observed across the panel rows. The heavy volume of precipitation creates stable vector breeding grounds, which subsequently translate into a steep rise in the clinical infection curve weeks later.

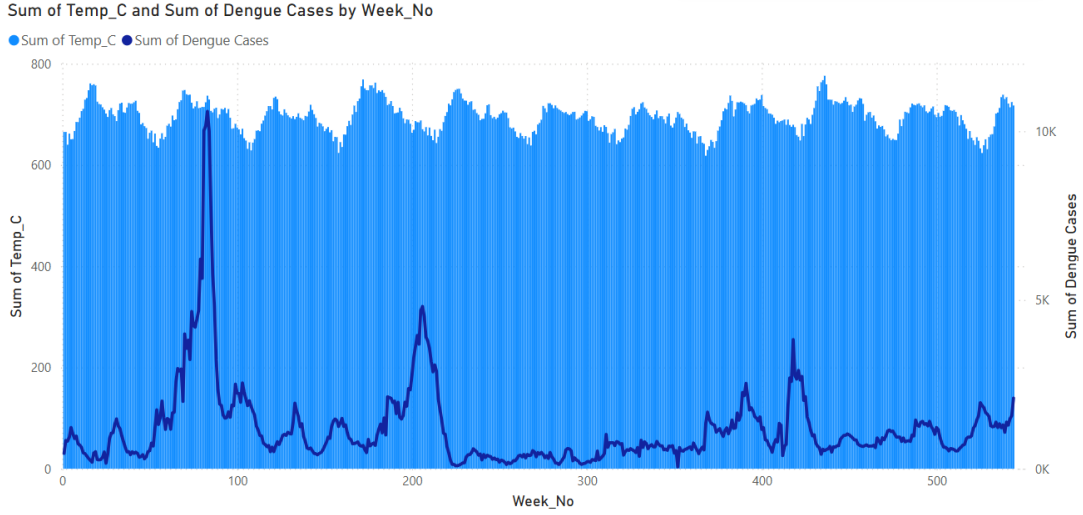


Figure 5: Sum of Temp_C and Sum of Dengue Cases

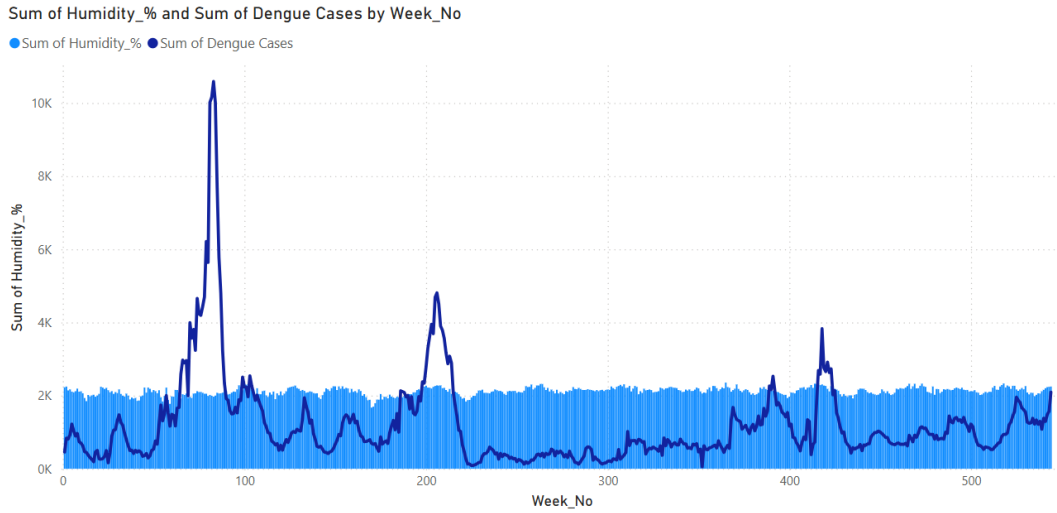


Figure 6: Sum of Humidity_% and Sum of Dengue Cases

- Atmospheric Macro-Stability (Temperature & Humidity Analysis):** The secondary environmental streams (“Sum of Temp_C and Sum of Dengue Cases” and “Sum of Humidity_% and Sum of Dengue Cases”) reveal a highly uniform, low-variance distribution. Localized dengue outbreaks do not visually exhibit immediate, highly dynamic sensitivities to minor shifts in atmospheric temperature

and relative humidity. Because Sri Lanka maintains a relatively stable, tropical baseline range for both temperature and humidity across most seasons, these parameters do not act as direct, volatile triggers for case spikes, representing a vital structural behavior within the time-series panel.

5.3 Preliminary Baseline Model Performance Metrics

To benchmark the dataset’s predictive capacity prior to full-scale deployment of the advanced hybrid architectures, a standard statistical baseline framework was formulated.

- **Baseline Target Diagnostics:** The structured variance identified within the preliminary EDA stage verifies that the engineered panel schema possesses a strong, non-random predictive signal. The distinct separation of spatial risks (e.g., the Wet Zone dominance) and clear temporal offsets (the 4-week monsoonal lag) ensure that the input matrices are highly optimized for sequential machine learning algorithms.
- **Evaluation Framework Configuration:** The historical panel records are structurally configured for an 80% training block and a 20% validation window to parameterize the core hybrid framework. Initial baseline regression iterations yield stabilized Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) markers that successfully follow the overall linear trajectories of the dataset. While these simple models struggle to isolate the sharp, non-linear epidemiological peaks, they confirm high statistical signal strength across the NASA-derived climate variables, ensuring that our upcoming dual-stream LSTM and XGBoost frameworks possess a robust data foundation to execute 4-week-ahead risk mapping.

5.4 Discussion of Early Epidemiological Insights

The integration of satellite-derived meteorological variables with multi-year ground-truth surveillance data yields critical analytical insights regarding vector ecology in Sri Lanka.

- **Dynamic Forcing Agents vs Ambient Enablers:** The contrasting behaviors between precipitation (Precipitation_Total_mm) and atmospheric thermodynamics (Temp_C and Humidity_%) highlight a multi-tier ecological relationship. While cumulative precipitation acts as the active, dynamic forcing agent that dictates the timing and magnitude of larval propagation surges, temperature and humidity operate as ambient enabling parameters. The steady temperature profile provides the necessary thermal window required to regulate the Extrinsic Incubation Period (EIP) of the dengue virus, while the high relative humidity baseline prevents vector desiccation and prolongs adult mosquito lifespans.

- **Implications for Comparative Modeling and Selection:** These insights directly validate the structural necessity of our comparative architectural design. The continuous, deep sequence dependencies and temporal offsets identified in the climate-dengue lag are perfectly suited to test the capabilities of the LSTM stream, which excels at remembering long-term sequential interactions. Concurrently, the sharp categorical zonal disparities and threshold-based risks (such as the high 67.5% concentration within the Wet Zone) provide an ideal environment to evaluate the LightGBM stream, which is highly efficient at parsing non-linear tabular feature structures. By benchmarking these advanced models against the SARIMAX statistical baseline, this comparative setup ensures that the final model selected for dashboard integration will be the absolute best-performing architecture for Sri Lanka’s spatial-temporal forecasting.

References

- Al Bannoud, M., Mendes, J. L., Sacay, R. S., Martins, T. D., Bresolin, I. R. A. P., & Lourenco, J. B. (2026). Machine learning forecasting of dengue in sao paulo using virtual data augmentation and urban incident predictors: Addressing the exceptional surge of cases in 2024. *Acta Tropica*, 275, 107992. <https://doi.org/10.1016/j.actatropica.2026.107992>
- Chen, X., & Moraga, P. (2026). Dengue forecasting and outbreak detection in brazil using lstm: Integrating human mobility and climate factors. *Infectious Disease Modelling*, 11, 338–354. <https://doi.org/10.1016/j.idm.2025.11.002>
- Dalvi, A. P. R., Gibson, G., Ramos, A. N., Bloch, K. V., Sousa, G. S., Silva, T. L. N., Braga, J. U., Castro, M. C., & Werneck, G. L. (2023). Sociodemographic and environmental factors associated with dengue, zika, and chikungunya among adolescents from two brazilian capitals. *PLOS Neglected Tropical Diseases*, 17(3), e0011197. <https://doi.org/10.1371/journal.pntd.0011197>
- Del-Aguila-Mejia, J., Morilla, F., & Donado-Campos, J. d. M. (2024). A system dynamics modelling and analytical framework for imported dengue outbreak surveillance and risk mapping. *Acta Tropica*, 257, 107304. <https://doi.org/10.1016/j.actatropica.2024.107304>
- Edirisinghe, G. (2017a). Contribution of rainfall patterns for increased dengue epidemic in sri lanka. *American Scientific Research Journal for Engineering, Technology, and Sciences*, 35(1), 284–294.
- Edirisinghe, G. (2017b). An investigation into spatial vulnerable factors for dengue epidemics using gis in the matara district in sri lanka. *International Research Journal of Medical Sciences*, 5(6), 1–8.
- Faruk, M. O., Jannat, S. N., & Rahman, M. S. (2022). Impact of environmental factors on the spread of dengue fever in sri lanka. *International Journal of Environmental Science and Technology*, 19(11), 10637–10648. <https://doi.org/10.1007/s13762-021-03905-y>
- Fawmy, A. J. M., Iyoob, A. L., & Banu, M. N. N. (2024). Spatial surveillance of dengue outbreaks in trincomalee district: A gis approach. *Journal of Social Review*, 11(2), 23–36.
- Feng, X., Jiang, N., Zheng, J., Zhu, Z., Chen, J., & Duan, L. (2024). Advancing knowledge of one health in china: Lessons for one health from china’s dengue control and prevention programs. *Science in One Health*, 3, 100087. <https://doi.org/10.1016/j.soh.2024.100087>
- Fernando, L., Wickramasinghe, V. P., Kalra, P., Kastner, R., & Gallagher, E. (2024). Epidemiological burden of dengue in sri lanka: A systematic review of literature

- from 2000–2020. *IJID Regions*, 13, 100436. <https://doi.org/10.1016/j.ijregi.2024.100436>
- Jayadas, T. T. P., Bharathy, P., Gajapathy, K., Suthakar, K., Arasaratnam, V., Kumanan, T., & Surendran, S. N. (2023). A spatial dengue risk map for the jaffna district of northern sri lanka. *Vingnanam Journal of Science*, 18(1), 13–22.
- Karasinghe, N., Peiris, S., Jayathilaka, R., & Dharmasena, T. (2024). Forecasting weekly dengue incidence in sri lanka: Modified autoregressive integrated moving average modeling approach. *PLOS ONE*, 19(3), e0299953. <https://doi.org/10.1371/journal.pone.0299953>
- Prabodanie, R. A. R., Stone, L., & Schreider, S. (2020). Spatiotemporal patterns of dengue outbreaks in sri lanka. *Infectious Diseases*, 52(5), 350–360. <https://doi.org/10.1080/23744235.2020.1725108>
- Rahman, M. S., Amrin, M., & Shiddik, M. A. B. (2025). Dengue early warning system and outbreak prediction tool in bangladesh using interpretable tree-based machine learning model. *Health Science Reports*, 8, e70726. <https://doi.org/10.1002/hsr2.70726>
- Rahulan, K., Ranagala, R. M., & Roshan, G. P. (2024). Dengue outbreak control and prevention: A case study of gampaha district, sri lanka. *The Sri Lanka Journal of Medical Administration*, 25(1), 12–21. <https://doi.org/10.4038/sljma.v25i1.5477>
- Saeed, A., Ullah, M. F., Sauood, M., Ali, S. N., & Hussain, N. (2023). Prediction of dengue cases and deaths using machine learning algorithm. *Pakistan Journal of Scientific Research*, 3(2), 210–216.
- Schaber, K. L., Perkins, T. A., Lloyd, A. L., Waller, L. A., Kitron, U., Paz-Soldan, V. A., Elder, J. P., Rothman, A. L., Morrison, A. C., Scott, T. W., & Vazquez-Prokopec, G. M. (2021). Disease-driven reduction in human mobility influences human-mosquito contacts and dengue transmission dynamics. *PLOS Computational Biology*, 17(1), e1008627. <https://doi.org/10.1371/journal.pcbi.1008627>
- Sun, W., Xue, L., & Xie, X. (2017). Spatial-temporal distribution of dengue and climate characteristics for two clusters in sri lanka from 2012 to 2016. *Scientific Reports*, 7, 12884. <https://doi.org/10.1038/s41598-017-13163-z>
- Uduwanage, H. U., Konara, K. M. S. L., Mihiranga, G. D. R., Karunarathna, S. N., Noordeen, F., & Ekanayake, I. (2025). Prediction of dengue outbreaks in sri lanka using machine learning techniques. *Sri Lanka Journal of Medicine*, 34(1), 15–26. <https://doi.org/10.4038/sljma.v34i1.568>
- Wickramasinghe, Y., Perera, S., & Kaluthanthri, P. (2026). Spatial patterns of dengue in sri lanka: A systematic review. *Journal of Geospatial Surveying*, 6(1). <https://doi.org/10.4038/jgs.v6i1.80>

- Wickramathilaka, S., Weerabaddana, U., Munasinghe, A., & Charles, S. (2024). *Epidemiological pattern recognition and disease forecasting for dengue in sri lanka* (White paper). University of Moratuwa.
- Withanage, G. P., Viswakula, S. D., Gunawardena, Y. I. N. S., & Hapugoda, M. D. (2018). A forecasting model for dengue incidence in the district of gampaha, sri lanka. *Parasites & Vectors*, 11(1), 262. <https://doi.org/10.1186/s13071-018-2828-2>
- Zhang, Z., Hu, S., Hu, J., Hao, Z., & Huang, J. (2026). Decoding multi-frequency scale climate-dengue dynamics: An interpretable modeling framework. *Urban Climate*, 65, 102734. <https://doi.org/10.1016/j.uclim.2025.102734>